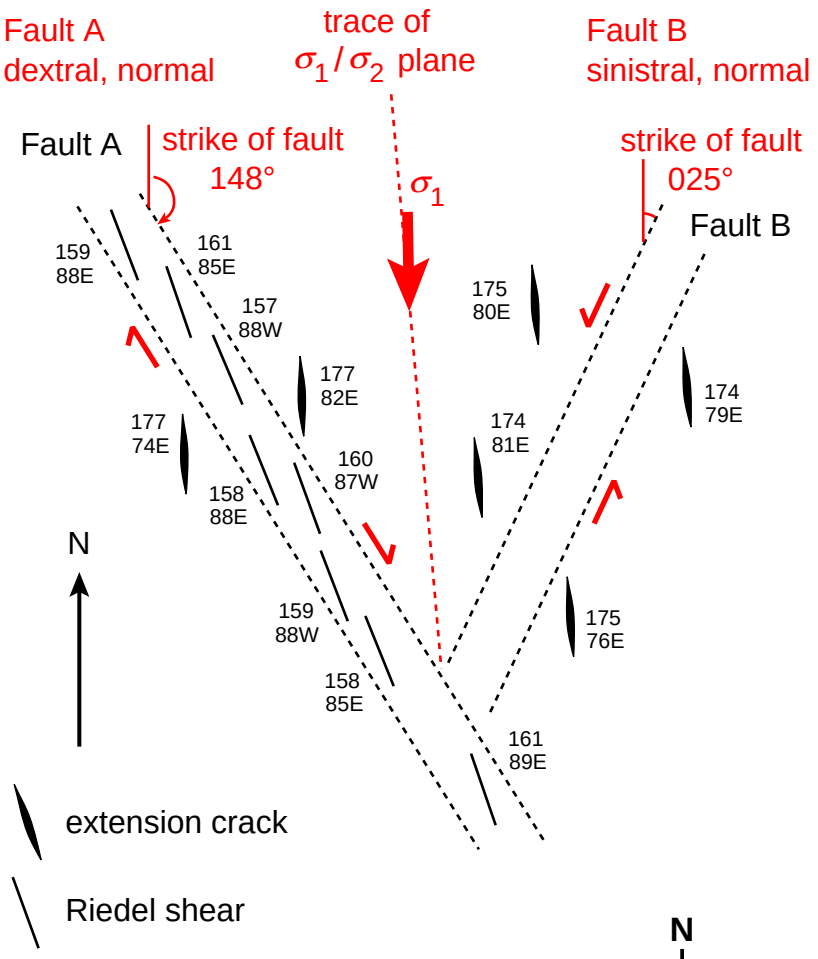
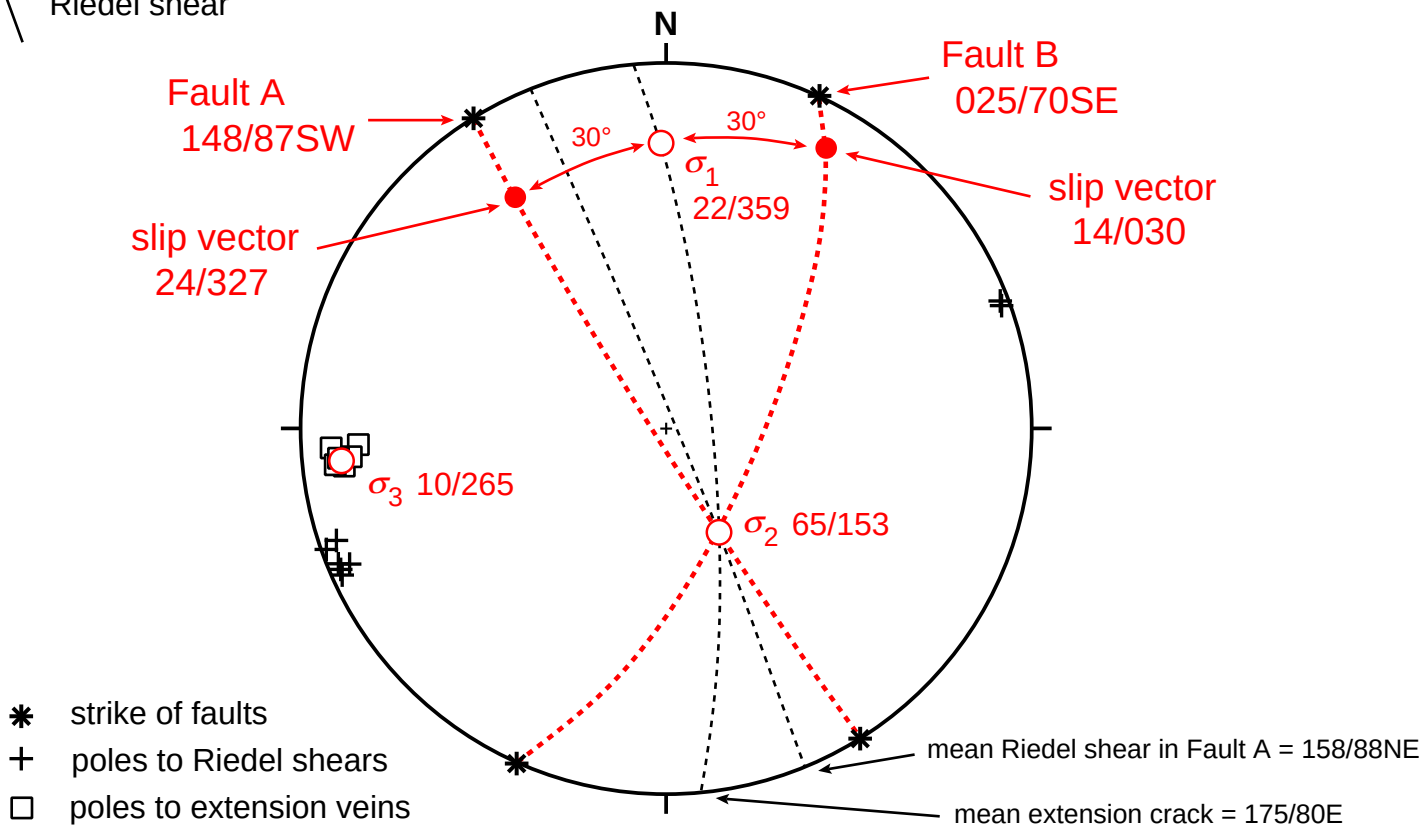
SHEAR ZONE EXERCISE

The map below shows the traces of two fault zones that form a conjugate pair. Reliable measurements of the fault planes could not be obtained but the orientations of Riedel shears and extension cracks were obtained. Using these data together with the observed traces on the ground, determine (a) the orientation of the two fault planes, (b) the orientation of the slip vector on each plane, (c) the sense of movement on each plane, and (d) the orientations of the principal stresses responsible for the formation of the faults



1. Plot poles to extension veins and poles to Riedel shears and choose an average value for each; draw a great circle for each of these average values − the intersection of these is the line of intersection of the two fault planes (= σ_2)
2. Measure the traces of each fault and use these, together with the line of intersection of the two fault planes to determine their orientation
3. Count around 90° on each fault plane from their line of intersection; these are the two slip vectors; the movement sense follows from the obliquity of the Riedel shears and extension veins
4. Draw in the great circle that has σ_2 as its pole; σ_1 lies at the intersection of this great circle with the great circle describing the average extension vein, and σ_3 is at 90° to σ_1 and σ_2